

Commutative Algebra & Algebraic Geometry

Exercise 10

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10.1 Problem

Let $n \in \mathbb{N}$, and denote by $\text{Aut}(\mathbb{P}^n)$ the group of automorphisms of $\mathbb{P}^n$.

- (1) Construct a homomorphism $\eta: \text{GL}_{n+1}(K) \rightarrow \text{Aut}(\mathbb{P}^n)$.
- (2) Prove that the center of $\text{GL}_{n+1}(K)$ consists of scalar matrices $K^\times I$.
- (3) Deduce that η factors as $\text{GL}_{n+1}(K) \rightarrow \text{PGL}_{n+1}(K) \xrightarrow{\bar{\eta}} \text{Aut}(\mathbb{P}^n)$ where $\bar{\eta}$ is injective.
- (4) Let $\mathcal{H}$ be the set of hyperplanes of $\mathbb{P}^n$. Show that $\text{Aut}(\mathbb{P}^n)$ acts transitively on $\mathcal{H}$.
- (5) Let $P_0, \dots, P_k \in \mathbb{P}^n$ be distinct points. Show that there is a hyperplane $H \in \mathcal{H}$ such that $P_0 \in H$ but $P_i \notin H$ for all other i .

- (1) Let $g \in \text{GL}_{n+1}(K)$, then g defines a map $\mathbb{A}^{n+1} - 0 \rightarrow \mathbb{A}^{n+1} - 0$ as it is injective. Since it is linear it acts on its coordinates linearly, thus defining a morphism. By problem 3 in exercise 9, this means it defines a unique morphism $\eta(g): \mathbb{P}^n \rightarrow \mathbb{P}^n$ such that $\pi \circ g = \eta(g) \circ \pi$. Then η is a homomorphism: for $g, h \in \text{GL}_{n+1}(K)$,

$$\eta(g \circ h) \circ \pi = \pi \circ (g \circ h) = (\pi \circ g) \circ h = \eta(g) \circ \pi \circ h = \eta(g) \circ \eta(h) \circ \pi.$$

By uniqueness of $\eta(g)$, this means that $\eta(g \circ h) = \eta(g) \circ \eta(h)$. Since $\eta(\text{id}) = \text{id}$, this means that η maps $\text{GL}_{n+1}(K)$ to automorphisms of $\mathbb{P}^n$ homomorphically.

- (2) If $g \in Z(\text{GL}_{n+1}(K))$, then g commutes with elementary matrices. In particular, for $c \in K^\times$ let $c\rho_i$ denote the diagonal matrix with ones on the diagonal except for c at the i th index. Then $(c\rho_i)g$ corresponds to the elementary row operation of multiplying g 's i th row by c , and $g(c\rho_i)$ is multiplying g 's i th column by c . Since $(c\rho_i)g = g(c\rho_i)$ this clearly means that g must be a diagonal matrix.

We claim that g is in fact a scalar matrix: so $g_{ii} = g_{jj}$ for all i, j . Let ρ_{ij} be the elementary matrix for swapping the i th and j th rows/columns. So $\rho_{ij}g$ is the matrix obtained by swapping g 's i th and j th rows, and $g\rho_{ij}$ is swapping g 's i th and j th columns. Then $\rho_{ij}g$'s element at the (i, j) th index is g_{jj} , while $g\rho_{ij}$'s is g_{ii} . Since $\rho_{ij}g = g\rho_{ij}$, this means $g_{jj} = g_{ii}$ as desired.

So all matrices in the center are scalar. The converse is clear, since multiplying by a scalar matrix (on either side) is the same as scaling the matrix.

- (3) We have $\eta(g) = \text{id}$ iff $\pi \circ g = \pi$, by uniqueness. This means that for every $P \in \mathbb{A}^{n+1} - 0$, $g(P)$ is a scalar multiple of P (since $\pi \circ g(P) = \pi(P)$). So all vectors are eigenvectors of g , which means that g must be scalar. Conversely a scalar matrix g clearly satisfies $\pi \circ g = \pi$.

Thus $\ker \eta = Z(\text{GL}_{n+1}(K))$, and so η factors as desired.

- (4) For a hyperplane $H = Z(\sum_i a_i x_i)$, H is the set of all $\pi(b_0, \dots, b_n)$ such that $\sum_i a_i b_i = 0$. For $g \in \text{GL}_{n+1}(K)$, let us conflate g and $\eta(g)$ so we can write $g \circ \pi = \pi \circ g$. Then

$$\pi(\bar{b}) \in gH \iff g^{-1}\pi(\bar{b}) \in H \iff \pi(g^{-1}\bar{b}) \in H \iff \bar{a} \cdot (g^{-1}\bar{b}) = 0,$$

and this in turn is iff $((g^\top)^{-1}\bar{a}) \cdot \bar{b} = 0$, which is iff $\bar{b}$ is in the hyperplane given by coefficients $(g^\top)^{-1}\bar{a}$. So $\text{GL}_{n+1}(K)$ action restricts to one on $\mathcal{H}$. Furthermore since $\text{GL}_{n+1}(K)$ acts transitively on $\mathbb{A}^{n+1} - 0$, it acts transitively on $\mathcal{H}$: for any $\bar{c}$ we can find $g \in \text{GL}_{n+1}(K)$ such that $(g^\top)^{-1}\bar{a} = \bar{c}$.

- (5) First let us assume $P_0 = [1 : 0 : \cdots : 0]$. Then a hyperplane H containing P_0 is any one of the form $H = Z(\sum_{i>0} a_i x_i)$ (i.e. $a_0 = 0$). Let $P_i = [b_{i,0} : \cdots : b_{i,n}]$; since $P_i \neq P_0$ we have that $b_{i,j} \neq 0$ for some $j > 0$. A hyperplane containing both P_0 and P_i is of the form

$$H = Z\left(\sum_{j>0} a_j x_j\right), \quad \sum_{i>0} a_j b_{i,j} = 0.$$

So there is a one-to-one correspondence between hyperplanes containing P_0 and P_i and vectors in the kernel of $\varphi_i: (a_1, \dots, a_n) \mapsto \sum_{j=1}^n a_j b_{i,j}$. Since the $b_{i,j}$ are not all zero, this kernel is not all of $\mathbb{A}^n$. Thus $\bigcup_i \ker \varphi_i$ is not all of $\mathbb{A}^n$ (since then it would be a vector space, and the finite union of vector spaces is a vector space iff one contains all the others, which would mean $\ker \varphi_i = \mathbb{A}^n$ for some i).

So let $(a_1, \dots, a_n)$ not in $\bigcup_i \ker \varphi_i$, then the hyperplane it defines $H = Z(\sum_{i=1}^n a_i x_i)$ contains P_0 but not contain any other P_i .

In general, since $\mathrm{GL}_{n+1}(K)$ acts transitively on $\mathbb{A}^{n+1} - 0$ and thus $\mathbb{P}^n$, there is a g such that $gP_0 = [1 : 0 : \cdots : 0]$. Then let H be such that $gP_0 \in H$ but $gP_i \notin H$ for all other i . Then $P_0 \in g^{-1}H$ but $P_i \notin g^{-1}H$ for all other i , so $g^{-1}H$ gives the desired hyperplane.

10.2 Problem

Let $n \geq 1$.

- (1) Prove that restriction forms an isomorphism $\mathcal{O}(\mathbb{A}^{n+1}) \rightarrow \mathcal{O}(\mathbb{A}^{n+1} - 0)$.
- (2) Deduce that $\mathcal{O}(\mathbb{P}^n) \cong k$ and thus that $\mathbb{P}^n$ is not an affine variety.

- (1) First we show that restriction is injective. Suppose that $f|_{\mathbb{A}^{n+1}-0} = 0$ for $f \in k[\mathbb{A}^{n+1}]$. Thus $Z(f) \supseteq \mathbb{A}^{n+1} - 0$ which is open, and thus dense since $\mathbb{A}^{n+1}$ is irreducible, meaning $Z(f) = \mathbb{A}^{n+1}$. Thus $f = 0$ as desired.

Since $\mathbb{A}^{n+1} - 0$ is open, its coordinate ring consists of rational functions f/g where g is nonzero on $\mathbb{A}^{n+1} - 0$. If g is nonzero on $\mathbb{A}^{n+1} - 0$, this means that $Z(g) \subseteq \{0\}$. But if g is not a scalar, then $Z(g)$ has dimension $n - 1$ and thus cannot be empty or a point; a contradiction. Thus g is a scalar, and f/g is a polynomial.

So sections of $\mathbb{A}^{n+1} - 0$ are polynomials, showing the surjectivity of restriction.

- (2) By definition

$$\mathcal{O}(\mathbb{P}^n) = \{f: \mathbb{P}^n \rightarrow k \mid f \circ \pi \in \mathcal{O}(\mathbb{A}^{n+1} - 0) = k[\mathbb{A}^{n+1}]\}.$$

Suppose $f \circ \pi = p \in k[\mathbb{A}^{n+1}]$, then this means for all $\lambda \in k^\times$

$$p(\lambda \bar{x}) = f \circ \pi(\lambda \bar{x}) = f \circ \pi(\bar{x}) = p(\bar{x}).$$

Writing $p = \sum_{\alpha} a_{\alpha} \bar{x}^{\alpha}$ this means that

$$\sum_{\alpha} \lambda^{|\alpha|} a_{\alpha} \bar{x}^{\alpha} = \sum_{\alpha} a_{\alpha} \bar{x}^{\alpha}.$$

Viewing this as a polynomial in λ , this means all of its non-constant coefficients are zero:

$$\sum_{|\alpha|=n} a_{\alpha} \bar{x}^{\alpha} = 0$$

for all $n > 0$. So for $|\alpha| > 0$ we have $a_{\alpha} = 0$. And thus p is a scalar.

Thus $f \circ \pi$ is a constant scalar, and so f is as well (by surjectivity of π). Thus $\mathcal{O}(\mathbb{P}^n) = k$ as desired.

If $\mathbb{P}^n$ were affine, its coordinate ring would be a quotient of $k[\mathbb{A}^m]$ for some m , by an ideal I and $\mathbb{P}^n \cong Z(I)$. But since its coordinate ring is k , I would have to be maximal, and then $Z(I)$ is a point. $\mathbb{P}^n$ is not a point, so it cannot be affine.

10.3 Problem

Prove that

- (1) being an isomorphism is local on the target,
- (2) being an isomorphism is not local on the source.

- (1) First we claim that if $\varphi: X \rightarrow Y$ is a morphism of SWFs and $U \subseteq Y$ is open, then the restriction $\varphi: \varphi^{-1}U \rightarrow U$ is a morphism of SWFs as well. This is clear since we are just restricting our perspective to open subsets of U and $\varphi^{-1}U$, which are open subsets of Y and X respectively.

Now if $\varphi: X \rightarrow Y$ is an isomorphism, let $U \subseteq Y$ be open. Then the restriction $\varphi: \varphi^{-1}U \rightarrow U$ remains a morphism with inverse $\varphi^{-1}: U \rightarrow \varphi^{-1}U$ (since this is just the restriction of the morphism φ^{-1}). Thus $\varphi|_{\varphi^{-1}U}$ is an isomorphism; in particular if φ is an isomorphism then it is locally an isomorphism on its target.

Conversely suppose $\{U_i\}$ is an open cover of Y such that $\varphi: \varphi^{-1}U_i \rightarrow U_i$ is an isomorphism for each i . Then φ is injective since if $\varphi x = \varphi y$, then these images lie in some U_i , so $x, y \in \varphi^{-1}U_i$ over which φ is bijective and thus $x = y$. And φ is surjective, as every element of U_i has a preimage in $\varphi^{-1}U_i$ since φ is an isomorphism there, and the U_i cover Y .

Thus φ^{-1} exists globally, and it is continuous since restricting it to the open cover $\varphi^{-1}U_i$ is continuous. It is also a morphism, since taking a basis of X by open subsets of $\varphi^{-1}U_i$ as i ranges, we see that φ^{-1} defines a morphism of coordinate rings from these basic open subsets, since φ^{-1} is a morphism $U_i \rightarrow \varphi^{-1}U_i$. By the equivalence of SWFs defined on basic open subsets and SWFs defined on all open subsets, we have the desired result.

- (2) Consider the morphism $\varphi = (\text{id}, \text{id}): \mathbb{A}^n \amalg \mathbb{A}^n \rightarrow \mathbb{A}^n$. Restricting φ to each copy of $\mathbb{A}^n$ in the source yields id , which is an isomorphism, but clearly φ itself is not as it is not even injective.

10.4 Problem

Prove that:

- (1) a composition of affine morphisms is affine,
- (2) an isomorphism of varieties is affine,
- (3) being an affine morphism is local on the target,
- (4) being an affine morphism is not local on the source.

- (1) Suppose $X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z$ are affine. Let $U \subseteq Z$ be open affine, then $\psi^{-1}U \subseteq Y$ is open affine as ψ is an affine morphism, and thus $(\psi \circ \varphi)^{-1}U = \varphi^{-1}(\psi^{-1}U)$ is affine as φ is affine. Thus $\psi \circ \varphi$ is affine.
- (2) Suppose $X \xrightarrow{\varphi} Y$ is an isomorphism, and let $U \subseteq Y$ be open affine. Then $\varphi: \varphi^{-1}U \rightarrow U$ is an isomorphism (being an isomorphism is local on the source), so $\varphi^{-1}U$ is isomorphic to an affine variety (as U is), and thus affine. Thus φ is affine.
- (3) Clearly if φ is affine, it is locally affine on the target; we just take a covering of the target by open affine subsets U_i , then $\varphi^{-1}U_i$ is affine by assumption. In recitation we showed that φ is affine iff there is an open cover of affine subsets $\{U_i\}$, $\varphi^{-1}U_i$ is affine. So conversely suppose there is an open cover of the target $\{W_i\}$ such that φ is affine $\varphi^{-1}W_i \rightarrow W_i$.

Since open subsets of algebraic varieties are algebraic varieties, we can cover each W_i by an open covering of affine subsets $\{U_i^j\}$, and then by assumption $\varphi|_{\varphi^{-1}W_i}(U_i^j) = \varphi^{-1}(U_i^j)$ is affine for each i, j .

Thus there is an open affine covering $\{U_i^j\}$ for which $\varphi^{-1}U_i^j$ is affine for all i, j . By the result from recitation, this means that φ is affine.

- (4) Consider $\varphi = (\text{id}, \text{id}): \mathbb{A}^1 \amalg \mathbb{A}^1 \rightarrow \mathbb{A}^1$. This is locally affine on its source, as it is locally an isomorphism on its source and isomorphisms are affine. But if φ were affine, this would imply that $\varphi^{-1}\mathbb{A}^1 = \mathbb{A}^1 \amalg \mathbb{A}^1$ is affine. The disjoint union has coordinate ring

$$\mathcal{O}(\mathbb{A}^1 \amalg \mathbb{A}^1) \cong \mathcal{O}(\mathbb{A}^1) \times \mathcal{O}(\mathbb{A}^1) \cong k[x] \times k[y].$$

Define $f: k[x, y] \rightarrow k[x] \times k[y]$ in the obvious manner, mapping x to $(x, 0)$ and y to $(0, y)$. This then maps

$$\sum_{n,k} a_{n,k} x^n y^k \mapsto \sum_{n,k} a_{n,k} (x^n, 0)(0, y^k) = \sum_n a_{n,0} (x^n, 0) + \sum_k a_{0,k} (0, y^k).$$

So its kernel is (xy) .

Since these are affine varieties, this means $\mathbb{A}^1 \amalg \mathbb{A}^1 \cong Z(xy) \subseteq \mathbb{A}^2$. The irreducible components of $Z(xy)$ are $Z(x), Z(y)$ (they are irreducible since their ideals are prime, and their union is $Z(xy)$). Since both copies of $\mathbb{A}^1$ are the irreducible components of $\mathbb{A}^1 \amalg \mathbb{A}^1$, this means that the isomorphism must map one copy to $Z(x)$ and the other to $Z(y)$. But $Z(x) \cap Z(y) = Z(x, y) = \{0\}$ is nonempty, so the isomorphism is not injective, a contradiction.

So φ is not affine, despite being locally affine on its source.

10.6 Problem

Let X be a topological space, show the following.

- (1) If $Y \subseteq X$ is a subspace, then $\dim Y \leq \dim X$.
- (2) If $X = \bigcup_i X_i$ is a finite union of closed subsets, then $\dim X = \max_i \dim X_i$.
- (3) If $X = \bigcup_i U_i$ is an open cover, $\dim X = \sup_i \dim U_i$.
- (4) Give an example of X and an open dense $U \subseteq X$ such that $\dim U < \dim X$.
- (5) Give an example of an infinite-dimensional Noetherian space.

- (1) Suppose $Y_0 \subseteq Y$ is closed irreducible, then we claim $\text{cl}_X(Y_0) \subseteq X$ is closed irreducible. Indeed, suppose $\text{cl}_X(Y_0) = Z_0 \cup Z_1$ are closed sets, then

$$Y_0 = \text{cl}_X(Y_0) \cap Y = (Z_0 \cap Y) \cup (Z_1 \cap Y),$$

where the first equality is because $\text{cl}_Y(Y_0) = \text{cl}_X(Y_0) \cap Y$ and Y_0 is closed. Since Y_0 is irreducible and $Z_i \cap Y$ are closed in Y , $Y_0 = Z_i \cap Y$ for some i . Thus

$$\text{cl}_X(Y_0) = \text{cl}_X(Z_i \cap Y) \subseteq \text{cl}_X(Z_i) = Z_i,$$

which means $\text{cl}_X(Y_0) = Z_i$, so $\text{cl}_X(Y_0)$ is indeed irreducible.

Now we claim that if $Y_0 \subsetneq Y_1 \subseteq Y$ are closed, then $\text{cl}_X(Y_0) \subsetneq \text{cl}_X(Y_1)$. Weak inclusion is clear. For strict inclusion let $y \in Y_1 - Y_0$, then $y \in \text{cl}_X(Y_1)$ but it cannot be in $\text{cl}_X(Y_0)$. If it were, then $y \in \text{cl}_X(Y_0) \cap Y = \text{cl}_Y(Y_0) = Y_0$ a contradiction.

Thus if $\emptyset \neq Y_0 \subsetneq \dots \subsetneq Y_n$ is a chain of irreducible closed subsets of Y , then $\emptyset \neq \overline{Y_0} \subsetneq \dots \subsetneq \overline{Y_n}$ is a chain of irreducible closed subsets of X . Thus $\dim Y \leq \dim X$ as desired.

- (2) By point (1), we have $\dim X_i \leq \dim X$ for all i and thus $\max_i \dim X_i \leq \dim X$. Now let $\emptyset \neq Z_0 \subsetneq \dots \subsetneq Z_n$ be a chain of irreducible closed subsets of X . We have

$$Z_n = \bigcup_i (Z_n \cap X_i)$$

and since Z_n is irreducible and this is a finite union of closed sets, there is some i where $Z_n = Z_n \cap X_i$, meaning $Z_n \subseteq X_i$. Thus this irreducible chain is contained in X_i ; thus $\dim X \leq \max_i \dim X_i$ as desired.

- (3) Again by point (1) we have $\sup_i \dim U_i \leq \dim X$. Now let $\emptyset \neq Z_0 \subsetneq \cdots \subsetneq Z_n$ be a chain of irreducible closed subsets of X . Let U_i be such that $Z_0 \cap U_i$ is nonempty (since Z_0 is nonempty and the U_i cover X , such an i exists). Since $Z_j \cap U_i$ is open in Z_j which is irreducible, and thus $Z_j \cap U_i$ is irreducible (open subsets of irreducible spaces are irreducible because irreducibility is equivalent to non-empty open subsets having non-empty intersection, if this is true for a space X it is clearly true for all open subsets of X).

Thus $\emptyset \neq Z_0 \cap U_i \subseteq \cdots \subseteq Z_n \cap U_i$ forms a chain of irreducible subsets of U_i , and they are all closed since the Z_j are. Furthermore, we claim $Z_j \cap U_i \subsetneq Z_{j+1} \cap U_i$. This is because $Z_{j+1} \cap U_i$ is dense in Z_{j+1} (nonempty open subsets of irreducible spaces are dense), but $Z_j \cap U_i$ is not: it is dense in Z_j but Z_j is closed and thus not dense in Z_{j+1} . So this is a proper chain of closed irreducible subsets of U_i , meaning $\dim X \leq \sup_i \dim U_i$ as desired.

- (4) Let $X = \{0, 1\}$ with the topology $\{\emptyset, X, \{0\}\}$; the closed subsets of X are then $\{\emptyset, X, \{1\}\}$. Then $\{0\} \subseteq X$ is open dense as the only closed subset it is contained in is X . But as a point, $\dim \{0\} = 0$.

On the other hand X is irreducible, since the only proper closed subset of X is $\{1\}$ and it is not equal to X . Thus $\emptyset \neq \{1\} \subsetneq X$ is a chain of closed irreducible subsets of X , so $\dim X = 1$. So $\dim \{0\} < \dim X$ despite $\{0\}$ being open dense in X .

- (5) Consider ω (the ordinal) equipped with the topology whose closed sets are $[n] = \{0, \dots, n-1\}$ for $n \leq \omega$ (setting $[0] = \emptyset$ and $[\omega] = \omega$). This is a topology as

$$[n] \cup [m] = [\max\{n, m\}], \quad \bigcap_i [n_i] = [\min_i n_i].$$

Furthermore this shows that each $[n]$ is irreducible, since if $[n] = [n_0] \cup [n_1]$ then $n = \max\{n_0, n_1\}$, so $n = n_0$ or $n = n_1$. Thus we have arbitrarily long chains of irreducible closed subsets

$$\emptyset \neq [1] \subsetneq [2] \subsetneq \cdots \subsetneq [n],$$

thus $\dim \omega = \infty$.

And ω is Noetherian: let Σ be a collection of closed subsets of ω . Let Σ_ω be the set of $n \leq \omega$ for which $[n] \in \Sigma$. Since ω is well-ordered, Σ_ω has a minimum, which means $[\min \Sigma_\omega] \in \Sigma$. Then

$$\bigcap_{[n] \in \Sigma} [n] = [\min \{n \leq \omega \mid [n] \in \Sigma\}] = [\min \Sigma_\omega] \in \Sigma.$$

Thus Σ has a minimal element; so ω 's closed subsets of the DCC and ω is Noetherian.